

Automatic Cat Feeder

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EENG 1910.001

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Abstract

This project involved many components of engineering design and teamwork. Our first steps involved coming up with an idea for a project proposal. We decided on a project that directly correlated to one of our group mates. They own a cat and we identified a problem of the cat needing the food twice a day but no one home to complete this task, so we decided to create a way to feed the cat on a timer based system to dispense food. Once this was completed we drew up a design for our automated cat feeder and then ordered the parts necessary for the system. We decided on creating a wheel-like structure that is turned by the power of a DC motor. It contains compartments when pushed over an opening they dispense food for the cat. The motor had its timing system dependent on code written in C++ which is put in an Arduino that reads the current time and if it's the correct time it then rotates the motor and completes the first feeding sequence. Next it completes the same sequence once the next time is proven true. This project taught our group how to make a project proposal and built on our previous knowledge and experience with work breakdown/allocation, resource allocation, technical writing, graph/chart construction, and how to come up with a solution to an identified problem.

Introduction

The problem we identified was that a person can not always be around for a cat's feeding time and rather than feeding them too soon, too late, or not at all; we decided on a way to time feeding that could be optimal for digestion, preference, and portion for any cat. The automated cat feeder we came up with was designed as an affordable and simple way to dispense cat food specialized to your cat's needs. This project has already been created and you may find products like this online ranging from lower end costing 20\$ and higher-end products costing 50-100\$. Ours, compared to others offers a more simple, affordable, and customizable option, being that the main design involves the motor and timer with you being able to change the compartment size and timing of feed times and portions based on container shape and size.

System Overview

The automatic cat feeder's system contains multiple electronic components designed to dispense food at different scheduled times throughout the day. The main components of the system consist of an Arduino Uno R3, a Real-Time Clock (RTC) Module, a DC motor, transistor, a diode, and a 9V battery power supply. The arduino is used as the microcontroller for the whole circuit. It stores the code that programs the RTC module so it reads the exact time it is at the giving moment. It also controls how long the DC motor should run for after the feeding time goes off. The RTC module is an integrated circuit (IC) that can read the date, time, and it has an alarm option. There is a backup coin cell battery at the back of the IC that turns into the main power supply. If there is a power failure in the main circuit, the RTC is still able to operate as usual. The DC motor being used in our circuit is a 3 rpm slow rotating motor that operates ideally with 12V. The amount of food that dispenses from the cat feeder is controlled by adjusting the amount of time the motor runs. An NPN (Negative-Positive-Negative) transistor is used in this circuit to act as the switch that allows the arduino to control the DC motor. The Arduino does not supply enough current to the motor, so the transistor allows a bit of current from the arduino to switch a larger current from the battery to the motor. The diode included in the circuit is used as a flyback diode. Its purpose is to protect the system from the DC motor's voltage spike. The diode is set up so that the polarity is opposite of the motor's polarity, which lets it so that it can only operate during the voltage spikes. When the motor turns off, it can produce reverse voltage and the diode redirects that current away from the sensitive components.

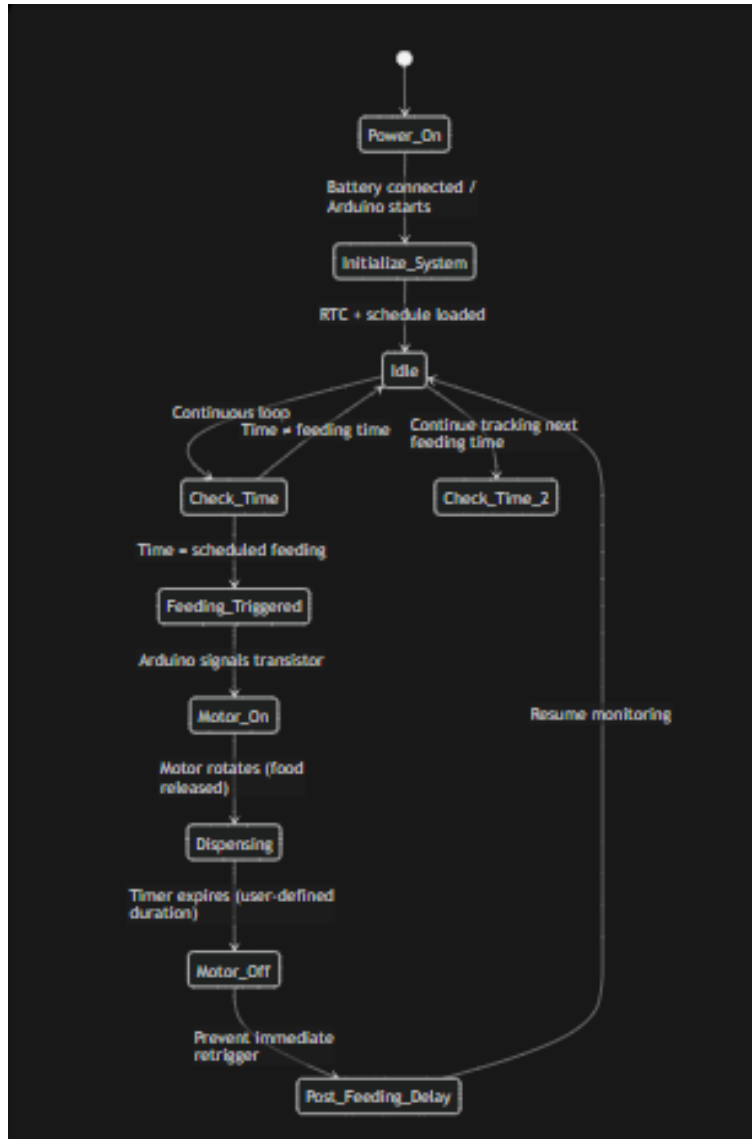


Figure 1.0: State diagram

Testing & Evaluation

The system was assembled following the schematic using a breadboard by connecting the Arduino, RTC module, transistor, diode, and DC motor. The power supply of the circuit is the 9V battery pack connected into the power and ground rails on the breadboard. A computer is used to program the Arduino so it can operate the entire system. The RTC module continuously keeps time, and the Arduino simultaneously compares them to the programmed feeding schedule. Once the time (i.e., 9:00 AM) occurs, the Arduino sends a signal to the transistor which acts like a switch to activate the motor. The program has a specific time interval given by the user expressing the duration the motor should run for. This ensures that the motor does not run endlessly causing all the food to dispense in one scheduled time. Once the first scheduled feeding

time occurs, the circuit keeps operating the same way, keeping track of time until the second feeding interval.

The system was first tested without the cat feeder prototype because we wanted to ensure the circuit alone works as intended. The current time is displayed in the computer's terminal and when the programmed time hits, the screen displays “Feeding Time” and the motor runs. This indicated that the circuit does work, but we still had to run a few more tests with the Arduino running independently from the computer. After programming, the Arduino was disconnected and now operated solely from the 9V battery pack. We tracked the time externally and awaited the feeding time. Once it hit the planned time, the motor started turning with a 5 second delay. The circuit did work which indicated to us that it was time to implement it with the prototype. With the motor integrated into the prototype, we ran multiple tests to ensure that both the mechanical and electrical systems were working well together.

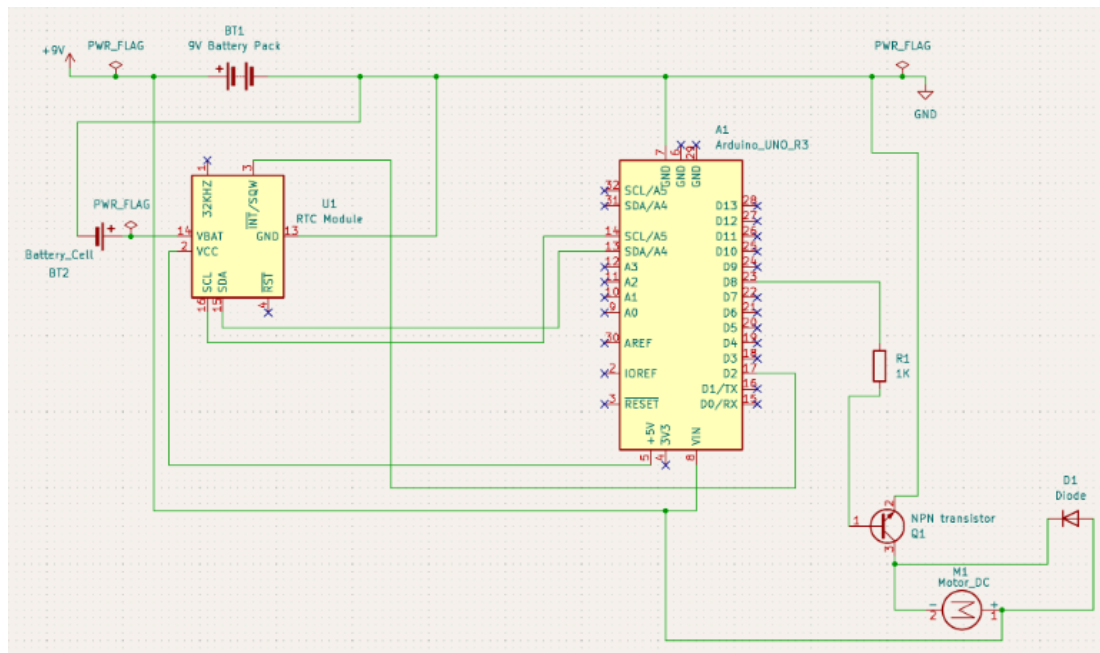


Figure 1.1: Schematic of The Cat Feeder

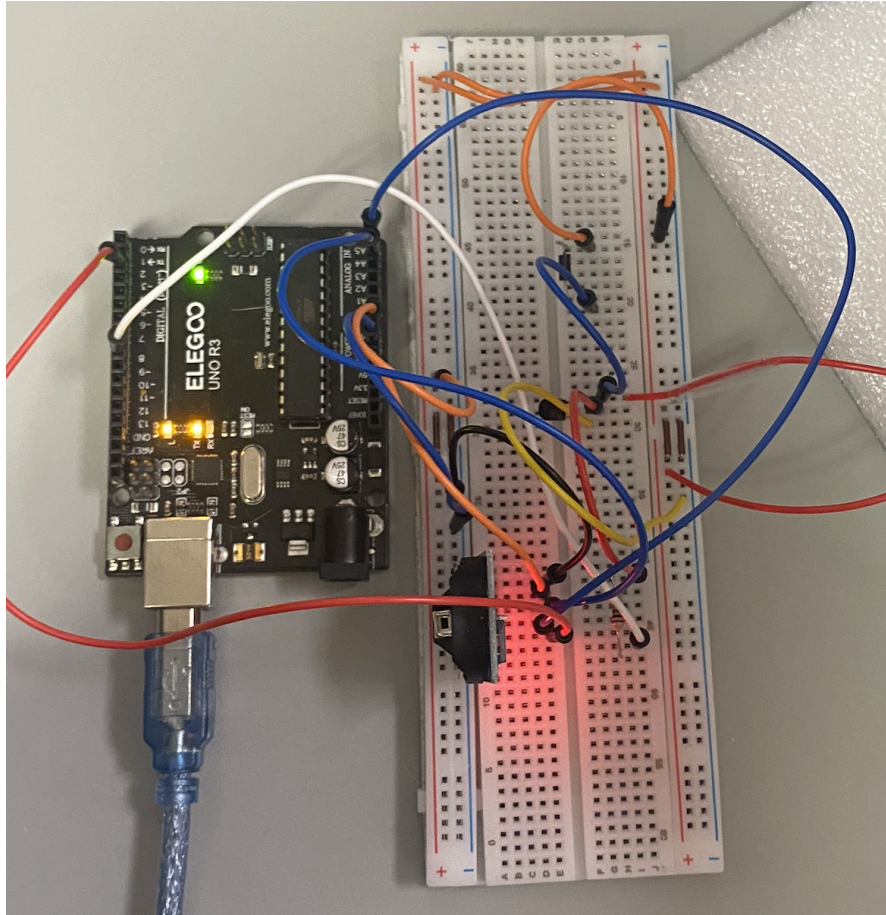


Figure 1.2 Cat feeder wiring

Results

Once a sufficient amount of tests were run, the system achieved its purpose of automatically dispensing food at different scheduled times. The RTC showed it could reliably timekeep and provide accurate time to the Arduino. The Arduino processed the time and triggered the feeding mechanism like it was programmed to. This showed that the system was running correctly. There is a minor issue involving the Arduino and RTC because there seems to always be a 5-7 second delay after the scheduled time hits. This seems to be inevitable in our prototype because when code is uploaded from the computer into the Arduino, it takes time to compile, transfer, and restart the new code. By the time the code is successfully transferred over, a few seconds have already passed. Even though this is an issue we face right now, the circuit still runs correctly within the time it is supposed to.

The DC motor runs as expected and it lasts the exact time the user programs it to run. For us, the time the motor runs is around 15-18 seconds. The time interval we put depends on how much cat food we put into each section of the wheel. This is because the heavier the food is, the slower the motor runs making us have to expand the time so the food all dispenses. Since this is considered the prototype for our automatic cat feeder, the battery pack is only 9V. The ideal power supply needed for the motor to run at 12 RPM is around ~12V. The power supply we have right now still provides enough power for our system to work, but later down the road replacing it with the 12V would be the best option.

During testing, there were several challenges we had to overcome. One of them being the motor had to be soldered with wires so we could plug it into the breadboard. After the wires were soldered, anytime we would move the motor around with the cat feeder, the metal part of the wires would break off. This made us have to desolder the broken part of the wire and solder new wires into the motor. This happened a few times because the wires we have are very fragile and tugging at them with excessive force would cause it to break off again. The temporary solution we came up with was to hot glue the newly soldered wires in the motor so it would be less likely for them to snap off again. This seemed to have fixed the issue temporarily, but will have to be re-done again in the future for a more practical way. Another issue that we faced was due to the mechanical side of our project, the cat feeder. Since the cat feeder was made using cardboard including the inside wall sections, there is food that becomes obstructed under the walls, having them get stuck and not turn anymore until the obstructed food is removed. This highlights that our cardboard feeder is a good temporary use to display that our system works, but not for permanent use. Despite all of our challenges, our system successfully operates and it shows that our concept of an automatic cat feeder works.

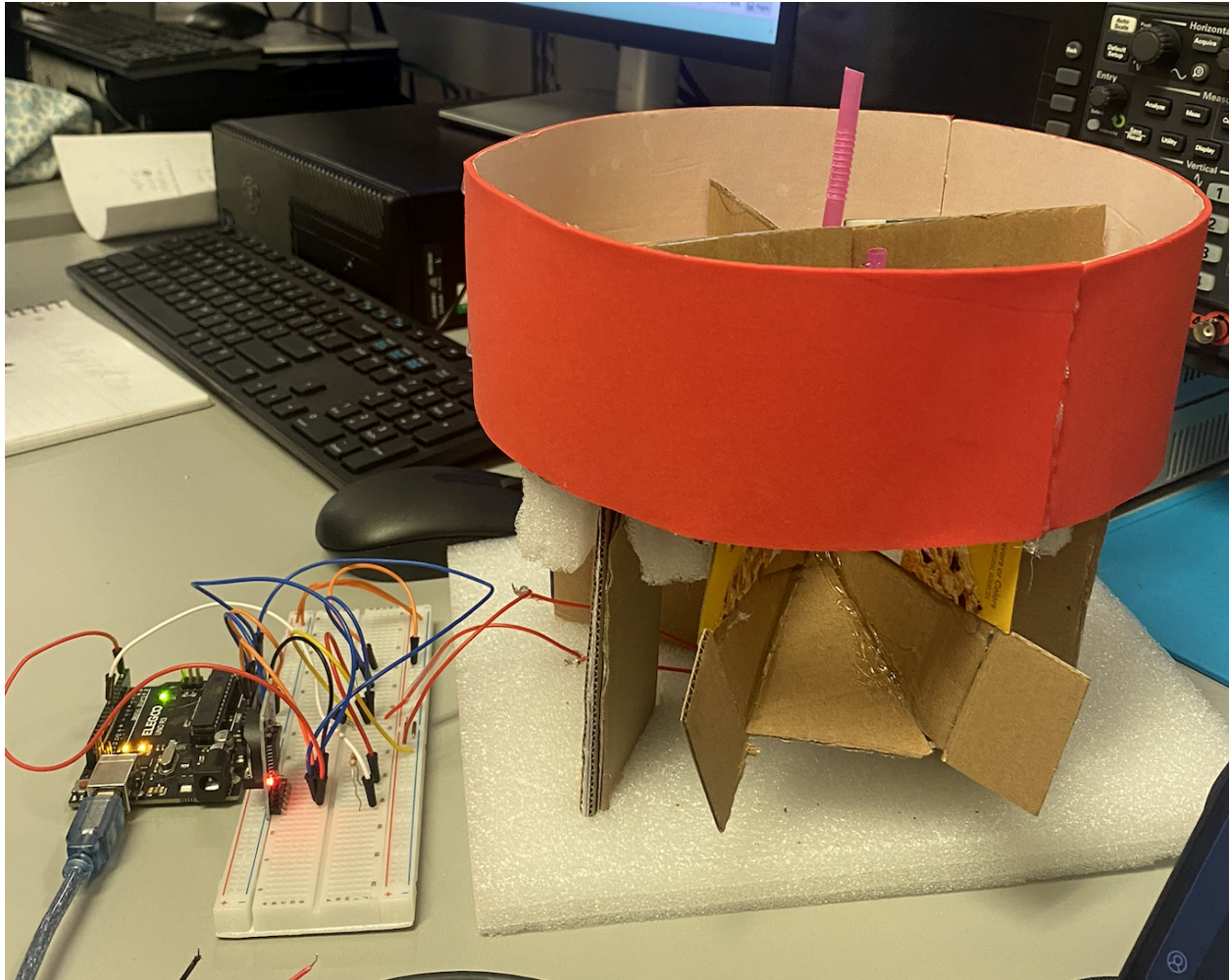


Figure 1.3 Final Version of the Automatic Cat Feeder

Project Management

Once assigned to our group we created a group chat to stay in consistent and available communication with each other, as well as a shared document used as a planner for the entire project, from design and problem identification to final code and product testing. We tried as best as possible to approach all steps together and divide tasks based on preference and competence. Our main goal was to maintain clear and efficient communication about each benchmark and problems faced throughout the engineering process. We learned that these were important techniques because it was very easy for us to be out of sync on different phases of the project due to different schedules, but with the help of our different platforms of collaboration we were able

to get past this issue. The most important and effective tool was that we created a proposal together allowing us to create a timeline and projection on how this project would transpire.

ID	Activity	Description	Time (max)	Check Point	People	Resources	Predecessors
1.0	Design Circuit	Complete the detailed design and Simulate on sim tools	7 Days		Ethan Yadira Jacob	Circuit sim tools	
1.1	Order/Gather Components	Buy and order components	7 Days		Ethan Yadira Jacob	Internet Shopping website for components	1.0
1.2	Code	Based on design develop program	3 Days	Rough draft code	Ethan Yadira Jacob	Coding software	1.0
1.3	Build Circuit	Assemble parts	1 Day	Circuit assembled	Ethan Yadira Jacob	Work bench DSPK Lab	1.1
1.4	Integrate Code & Circuit	Integrate code with circuit	4 Days		Ethan Yadira Jacob	Work bench DSPK Lab and PC	1.2 & 1.3
1.5	Troubleshoot	Look at code and circuit see if it runs	1 Day		Ethan Yadira Jacob	PC	1.4
1.6	Document Issues	Document issues and errors	1 Day		Ethan Yadira Jacob		1.5
1.7	Refine and Fix (Final Product)	Finalize code and Circuit with correcting previous issues	2 Days	Final code and circuit	Ethan Yadira Jacob	Coding software PC Circuit sim tools	1.6

Figure 1.4: Timeline of project

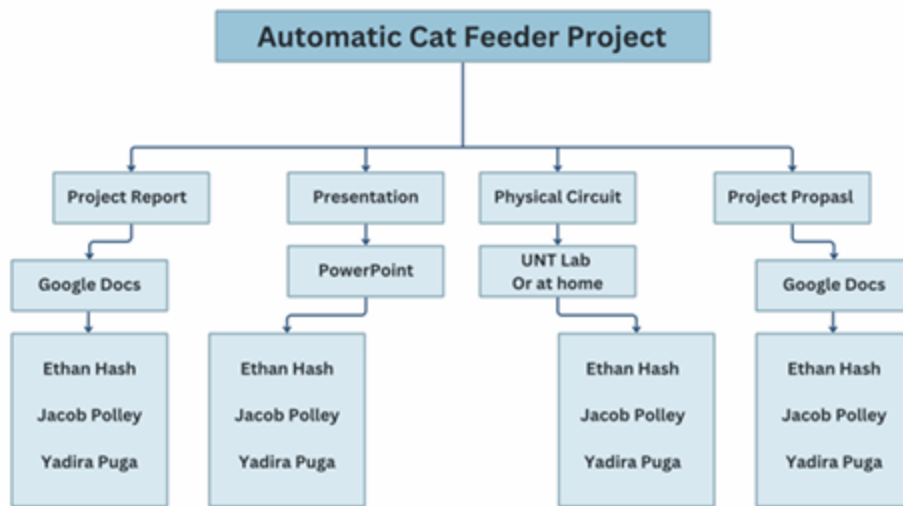


Figure 1.5: Work Breakdown chart

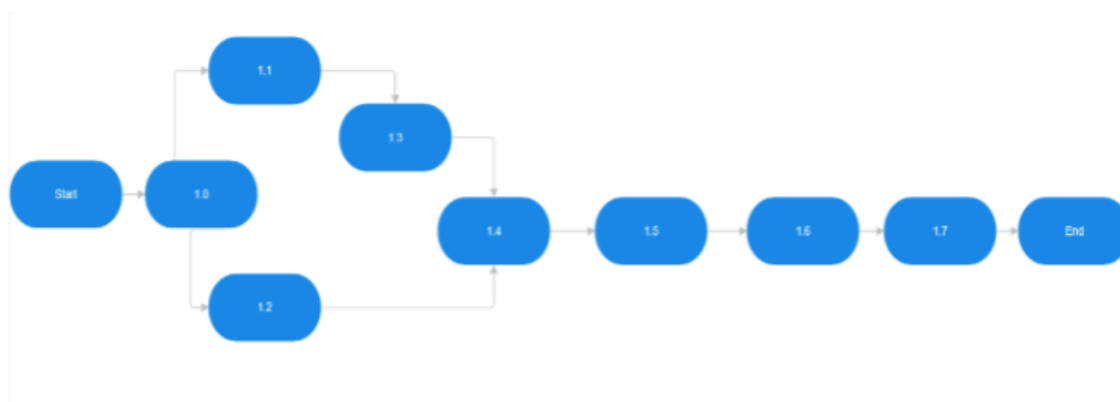


Figure 1.6: Network Diagram

Ethical & Legal Issues

With this product there could be some legal or ethical issues that stem from design and implementation in a home or setting. One may be we have not created a users guide or safety manual to instruct any boundaries or guidelines of timing, power input, or application of the feeder. This could lead to users misusing the product or having issues that could be resolved by a manual. Another large issue could be that it has not undergone any mass production or intensive testing. This could lead to unknown errors or dangers not yet discovered because of limited use and testing.

Future Improvements

With the limited time for this project we were able to create a product that solves the identified problem and has some benefits compared to other models. This is definitely not a perfect product but is something that could definitely have many different

improvements. Adding an easier way to manipulate timing and power to the motor based on load of food could be a great addition. This could allow a more user-friendly way to choose feeding time regardless of coding background and way to help the motor based on cat food weight; some people may be using the product to feed more than one cat. Another improvement would be the implementation of a button or sensor based on preference of control over the RTC module. This would be a more universal way to control the RTC module rather than the use of a computer or code.

Conclusion

In conclusion our automatic cat feeder project was successful in solving the problem we originally identified. We were able to design and build a system that uses an Arduino, RTC module, and DC motor to dispense food at scheduled times throughout the day. Through testing we confirmed that the system works as intended with only minor issues such as small timing delays and mechanical limitations from the prototype materials. This project helped us apply what we've learned about circuits, programming, and system design in a real-world situation, while also improving our teamwork and communication skills. Although there are still improvements that can be made, such as strengthening the structure and making the system more user-friendly, our final product shows that our idea is practical and effective.

Citations

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